SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What are the optical detectors used in optical communication. Explain the principle and operation of PIN diode.

Q.24 What are the optical amplifiers used in optical communication. Explain the principle and operation of EDFA.

Q.25 Explain the different types of optical fiber losses and how to reduce its effects.

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Roll No. ....

5th Sem/

ECE, ECE (For Speech & Hearing Impaired)

Subject : Optical Fibre Communication

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 A LED is a

a) Coherent source of light

b) Non Coherent source of light

c) Best source of light

d) None

Q.2 An optical fiber is made up of _____ material.

a) Glass b) Rubber

c) Plastic d) Copper

Q.3 The bandwidth in fiber optical communication is represented in terms of ______.

a) Wavelength b) Frequency

c) Both (a) and (b) d) None of these

Q.4 In a single mode of optical fiber_____.

a) Bandwidth is higher (1000 MHz)

b) Long communication

c) V is less than 2.402

d) All of these

Q.5 Optical amplifiers can support _____.

a) Time division multiplexing

b) Wave division multiplexing

c) Both (a) and (b)

d) None of these

Q.6 How many types of techniques are there in fiber splicing?

a) One b) Two

c) Three d) Four

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write the full form ILD.

Q.8 Write the full form APD.

Q.9 Write the full form SOA.

Q.10 What is critical angle?

Q.11 Define photodetector.

Q.12 Write the full form SMF.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write the comparison between LED and ILD.

Q.14 Write the characteristics of APD.

Q.15 What are the effects of dispersion in optical communication?

Q.16 Write the characteristics of ILD.

Q.17 Write the comparison between step indexed and graded indexed fiber.

Q.18 Write the principle of light penetration.

Q.19 Write the short note on construction details of optical fiber.

Q.20 Write the short note on scattering losses in optical fiber.

Q.21 Write a short note on splicing techniques.

Q.22 Write a short note on absorption losses.